

1 Solution — GIAM 7.4

Problem 4

1.1 Problem

The textbook displays a **3**-dimensional analog of Pascal's triangle — *Pascal's tetrahedron* — through layer $n = 3$. What would the next layer (layer $n = 4$) look like?

1.2 What Pascal's Tetrahedron Encodes

Just as Pascal's triangle holds the binomial coefficients $\binom{n}{k}$ — the coefficients of $(x + y)^n$ — Pascal's tetrahedron holds the **trinomial coefficients**

$$\binom{n}{i, j, k} := \frac{n!}{i! j! k!}, \quad i, j, k \geq 0, \quad i + j + k = n$$

— the coefficients of

$$(x + y + z)^n = \sum_{i+j+k=n} \binom{n}{i, j, k} x^i y^j z^k.$$

The layer- n slice is a triangular array of $\binom{n+2}{2} = \frac{(n+1)(n+2)}{2}$ entries, one for each non-negative triple (i, j, k) summing to n . The three corners of the triangle correspond to the “pure” triples $(n, 0, 0), (0, n, 0), (0, 0, n)$ — each carrying the entry $\binom{n}{n, 0, 0} = 1$.

Per-layer entry counts match the diagram in the problem:

Layer n	# entries = $\binom{n+2}{2}$	Entries
0	1	1
1	3	1, 1, 1
2	6	1, 2, 1, 2, 2, 1 (three 1 \$s on corners, three 2 \$s on edges)

3	10	three 1 \$s, six 3 \$s, one 6
4	15	three 1 \$s, six 4 \$s, three 6 \$s, three 12 \$s

Table 1.1

1.3 The 3D Pascal Recursion

Pascal's familiar 2D rule "*every entry is the sum of the two entries above it*" generalizes to *three* parents per entry in the tetrahedron:

$$\binom{n}{i, j, k} = \binom{n-1}{i-1, j, k} + \binom{n-1}{i, j-1, k} + \binom{n-1}{i, j, k-1},$$

with the convention that a coefficient with any negative index is 0. This is the trinomial analog of Pascal's identity, and it follows from a direct calculation on factorials (or, more conceptually, from the recurrence $(x+y+z)^n = (x+y+z) \cdot (x+y+z)^{n-1}$, distributed by which of x, y, z is chosen as the new factor).

1.4 Layer 4 — The Answer

Applying the recursion to every triple (i, j, k) with $i+j+k=4$ yields the 15 trinomial coefficients of layer 4:

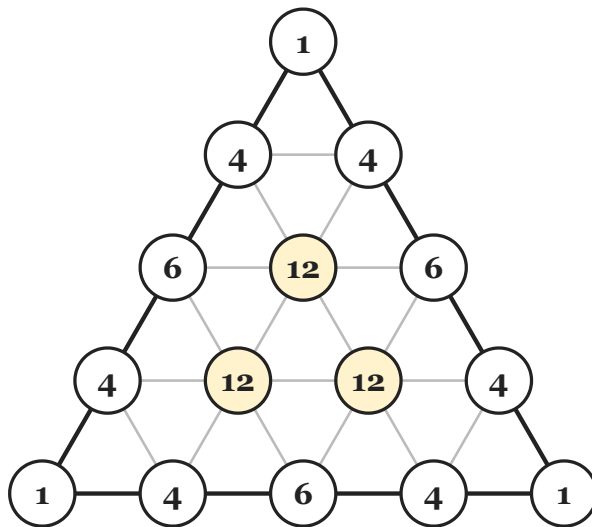


Figure 1.1: The next layer of Pascal's tetrahedron is a triangular array with 15 entries. The three corner entries are 1; each of the three edges carries the values 1, 4, 6, 4, 1 (Pascal's row 4); and the three interior entries (highlighted) are all 12.

Explicitly:

$$\begin{array}{ccccc}
 & & 1 & & \\
 & 4 & & 4 & \\
 & 6 & 12 & 6 & \\
 4 & 12 & 12 & 12 & 4 \\
 1 & 4 & 6 & 4 & 1
 \end{array}$$

1.5 Verification — Recursion-by-Recursion

Sample checks of the 3D Pascal rule applied to layer $\mathbf{3} \rightarrow$ layer $\mathbf{4}$:

Corner $(4, 0, 0) = 1$. Parents: $\binom{3}{3,0,0} + \binom{3}{4,-1,0} + \binom{3}{4,0,-1} = 1 + 0 + 0 = 1$. ✓

Edge near-corner $(3, 1, 0) = 4$. Parents: $\binom{3}{2,1,0} + \binom{3}{3,0,0} + \binom{3}{3,1,-1} = 3 + 1 + 0 = 4$. ✓

Edge midpoint $(2, 2, 0) = 6$. Parents: $\binom{3}{1,2,0} + \binom{3}{2,1,0} + \binom{3}{2,2,-1} = 3 + 3 + 0 = 6$. ✓

Interior $(2, 1, 1) = 12$. Parents: $\binom{3}{1,1,1} + \binom{3}{2,0,1} + \binom{3}{2,1,0} = 6 + 3 + 3 = 12$. ✓

The same recipe yields every entry; symmetry under permutations of (i, j, k) reduces the work to four distinct cases (corner, near-corner edge, edge midpoint, interior).

1.6 Verification — Sum of Entries

By the binomial-like identity

$$3^n = (1 + 1 + 1)^n = \sum_{i+j+k=n} \binom{n}{i, j, k},$$

the sum of all entries in layer n must equal 3^n . For layer $\mathbf{4}$:

$$3 \cdot 1 + 6 \cdot 4 + 3 \cdot 6 + 3 \cdot 12 = 3 + 24 + 18 + 36 = 81 = 3^4. \checkmark$$

1.7 Structure of Layer 4

Three observations about how the 15 entries of layer 4 organize themselves:

1.7.1 1. The Three Edges Carry Pascal's Row 4

Each edge of the triangle consists of 5 entries where one of the three indices is 0. For the edge with $k = 0$:

$$\binom{4}{4,0,0}, \binom{4}{3,1,0}, \binom{4}{2,2,0}, \binom{4}{1,3,0}, \binom{4}{0,4,0} = 1, 4, 6, 4, 1.$$

But $\binom{4}{i,j,0} = \binom{4}{i,4-i,0} = \frac{4!}{i!(4-i)!0!} = \binom{4}{i}$, the ordinary binomial coefficient. So each edge of layer 4 is *literally* row 4 of (ordinary) Pascal's triangle, 1, 4, 6, 4, 1. By symmetry, the same holds for the other two edges (with $i = 0$ or $j = 0$).

1.7.2 2. The Three Corners Are Always 1

The corner entries correspond to $\binom{n}{n,0,0} = \binom{n}{0,n,0} = \binom{n}{0,0,n} = 1$ — these are the coefficients of x^n, y^n, z^n in $(x + y + z)^n$, each appearing exactly once.

1.7.3 3. The Three Interior Entries Are All 12

Layer 4 has three positions where *all three* of i, j, k are positive:

$$(2, 1, 1), \quad (1, 2, 1), \quad (1, 1, 2).$$

The trinomial coefficients are invariant under permutations of (i, j, k) , so all three carry the same value $\binom{4}{2,1,1} = \frac{4!}{2!1!1!} = 12$. Together they form a small inner triangle of 12 \$s — the "interior" of layer 4 \$.

1.8 Remark — Permutation Symmetry (D_3)

Because $\binom{n}{i,j,k}$ depends only on the *multiset* $\{i, j, k\}$ (not the order), every layer of Pascal's tetrahedron has *dihedral D_3 symmetry*: it is invariant under the **6** permutations of $\{x, y, z\}$ (three rotations by 120° around the triangle's center, and three reflections across the medians).

Equivalence classes under D_3 :

Class	Triples (up to permutation)	Coefficient	Multiplicity in layer 4
All identical (singleton)	—	—	0 (no (a, a, a) with $3a = 4$)
Two equal, one different (pair)	$\{4, 0, 0\}$	1	3
”	$\{2, 2, 0\}$	6	3
”	$\{2, 1, 1\}$	12	3
All distinct	$\{3, 1, 0\}$	4	6

Table 1.2

Total: $3 + 3 + 3 + 6 = 15$. ✓

This explains the entry count for free: each *unordered* triple contributes **3** entries (if exactly two are equal) or **6** entries (if all distinct) or **1** entry (if all three are equal, which doesn't happen for $n = 4$).

1.9 Remark — Counting Layers in a Tetrahedron

Stacking layers $0, 1, 2, \dots, N$ gives the *first* $N + 1$ layers of Pascal's tetrahedron, which together hold

$$\sum_{n=0}^N \binom{n+2}{2} = \binom{N+3}{3}$$

entries — a *tetrahedral number*. For $N = 4$, the total is $\binom{7}{3} = 35$ entries across all layers 0 through 4.

Through layer n	Total entries (tetrahedral number)
0	1
1	4
2	10
3	20
4	35

Table 1.3

So adding layer 4 contributes $35 - 20 = 15$ new entries, consistent with our count.

1.10 Remark — Higher Dimensions

The pattern continues: Pascal's *simplex* in d dimensions holds the *multinomial* coefficients of $(x_1 + x_2 + \cdots + x_d)^n$. The recursion has d parents per entry:

$$\binom{n}{i_1, \dots, i_d} = \sum_{r=1}^d \binom{n-1}{i_1, \dots, i_r-1, \dots, i_d}.$$

For $d = 2$: ordinary Pascal's triangle (2 parents per entry). For $d = 3$: Pascal's tetrahedron (3 parents per entry) — this problem. For $d = 4$: Pascal's *pentatope* (4 parents per entry).

Each layer of the d -simplex has $\binom{n+d-1}{d-1}$ entries, generalizing the $\binom{n+2}{2}$ count for the tetrahedron.